grab_samples.csv

Grab samples include both surface water (stream sites S1–S17.b) and groundwater (well sites WC7, WD5–6, WE1–6, WF3–4, WG1–2, WH3–5). Sampling dates span June 2024–November 2025. Data completeness varies among sampling dates and sites because not all constituents were measured during every sampling event. Several constituents, including EEM/DOM optical properties, were measured only during the August 2025 synoptic sampling event, while nutrients, major ions, stable isotopes, and radon were measured on varying subsets of dates and sites.

Column	Description
Date	Sample collection date
site_position	Site identifier (S prefix = surface water; W prefix = groundwater)
Distance_m	Distance upstream from gage house (m); NA for groundwater wells
water_type	Water source classification: Surface or Groundwater
ALK_mg_L	Alkalinity (mg/L)
DIC_mg_L	Dissolved inorganic carbon (mg/L)
DOC_mg_L	Dissolved organic carbon (mg/L)
NH3_mg_L	Ammonia (mg/L)
NO3_mg_L	Nitrate (mg/L)
PO4_mg_L	Orthophosphate (mg/L)
CA_mg_L	Calcium (mg/L)
Cl_mg_L	Chloride (mg/L)
K_mg_L	Potassium (mg/L)
MG_mg_L	Magnesium (mg/L)
SO4_mg_L	Sulfate (mg/L)
TN_mg_L	Total nitrogen (mg/L)
TP_mg_L	Total phosphorus (mg/L)
NA_mg_L	Sodium (mg/L)
dD_avg	Average deuterium (δ^2H; ‰ vs. VSMOW)
dD_sd	Deuterium standard deviation (‰)
d18O_avg	Average oxygen-18 ($\delta^{18}O$; ‰ vs. VSMOW)
d18O_sd	Oxygen-18 standard deviation (‰)
Dexcess	Deuterium excess (‰; $\delta^2H - 8 \times \delta^{18}O$)
Radon_pCi_L	Radon concentration (pCi/L)
abs254	Absorbance at 254 nm; used with DOC concentration to calculate SUVA254
int_abs	Integrated absorbance from 300–400 nm
slope_275_295	Absorbance spectral slope from 275–295 nm
slope_350_400	Absorbance spectral slope from 350–400 nm
slope_ratio	Spectral slope ratio
slope_300_700	Absorbance spectral slope from 300–700 nm

Column	Description
SUVA254	Specific UV absorbance at 254 nm, calculated by normalizing absorbance at 254 nm by DOC concentration (L mg C^{-1} m^{-1})
flour_index	Fluorescence index; summary metric describing relative fluorescence at selected wavelengths
peakA	Peak A fluorescence; relative abundance of a characteristic fluorescent DOM component
peakC	Peak C fluorescence; relative abundance of a characteristic fluorescent DOM component
peakT	Peak T fluorescence; relative abundance of a characteristic fluorescent DOM component
int_flour	Integrated fluorescence; broad measure of total fluorescence associated with DOM